

Recommendations — pancreatic-mets-kras-g12d-basal-arid1a -k9r3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — pancreatic-mets-kras-g12d-basal-arid1a-k9

Profile snapshot

- **Primary site:** pancreas
- **Histology:** pancreatic ductal adenocarcinoma (PDAC); PurlST RNA signature classifies the tumor as the basal-like transcriptional subtype
- **Stage:** IV (metastatic): large primary pancreatic mass with multiple liver metastases each >1 cm
- **ECOG:** 1
- **Age band:** 70-79
- **Sex:** male
- **Biomarkers:** KRAS G12D mutated; MLH1 mutation somatic mutation reported (single MMR-gene variant; tumor is microsatellite stable and germline-negative); SMAD4 E520 *stop-gain (predicted loss of function)*; TP53 Y126S and A138V (two missense variants); ARID1A frameshift (predicted loss of function); TMB 8.6 mutations/Mb; Microsatellite status MSS (microsatellite stable); PurlST subtype basal-like; Germline status no germline variants detected; Gene rearrangements none detected; HLA type A01:01, A02:01, B37:01, B44:02, C05:01, C*06:02; Additional xR variants ABCC3 missense; ASXL1 frameshift; BCL7A splice-region; BCLAF1 stop-gain; LRP1B splice-region
- **Efficacy/toxicity weight:** 0.5

4 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Kras G12D interventions

1. daraxonrasib (RMC-6236), pan-RAS(ON) inhibitor — front-line via RASolute 303 (NCT07491445)

Likelihood of effect: High in 2L RAS G12 PDAC — OS HR 0.40 (95% CI 0.30-0.53) in RASolute 302 (n=500); first-line benefit inferred from the unread RASolute 303.

Toxicity burden: Moderate (rash G3+ 14%, stomatitis G3+ 12%; G3+ TRAE 43.6% overall, below chemo)

Counter-productive MoA: Low (Mucocutaneous toxicity can force dose reductions that erode the exposure the survival signal depends on; no mechanism-level antagonism of the therapeutic goal.)

Overall: The only feature-targeting option with randomized phase 3 survival evidence; first-line benefit inferred from an unread trial and the allele estimate borrowed from a RAS G12 subgroup.

Key references: PMID 42223072 · NCT07491445 · NCT06445062

2. zoldonrasib (RMC-9805), KRAS G12D(ON)-selective inhibitor — front-line via RASolute 305 (NCT07621718)

Likelihood of effect: Moderate, allele-matched — ORR 30% / DCR 80% in G12D PDAC (phase 1, n=40, no CI;

PFS/DoR immature). RASolute 305 will test it front-line.

Toxicity burden: Low (nausea 27%, diarrhea 20%, rash 10%; no G4/5 events at monotherapy dose)

Counter-productive MoA: **Moderate** (Critic and consensus site dissented on the thin single-arm evidence base; durability unmeasured, so on-target activity may not translate to a durable effect.)

Overall: The allele-matched, gentlest agent in the set — best tolerability for a 79-year-old, but the evidence is a single-arm n=40 abstract with no durability readout.

Key references: doi:10.1200/JCO.2025.43.4_suppl.724 · NCT07621718 · NCT06040541

3. INCB161734, KRAS G12D-selective inhibitor — front-line + chemo via DAWN-303 (NCT07522073)

Likelihood of effect: Moderate to high single-agent — ORR 37% / DCR 78% in pretreated G12D PDAC (phase 1, n=41, no CI; PFS not reached at ~5 mo). DAWN-303 tests it front-line.

Toxicity burden: High (combination-cohort G3+ neutropenia ~40%; monotherapy GI 50-60% grade 1-2)

Counter-productive MoA: **Moderate** (Conservative dissented on additive marrow toxicity: chemo-combination neutropenia can force dose reductions in an older patient that erode delivered intensity.)

Overall: Strongest single-agent response number in the dossier, but the enrollable front-line construct bundles chemotherapy with ~40% grade ≥3 neutropenia and no dedicated combination-safety data.

Key references: doi:10.1200/JCO.2026.44.2_suppl.654 · NCT07522073 · NCT06179160

Evidence in detail

daraxonrasib (RMC-6236), pan-RAS(ON) inhibitor — front-line

Wolpin et al. *N Engl J Med* 2026

- **Disease context:** previously treated metastatic RAS-mutant pancreatic ductal adenocarcinoma (2L+) — metastatic PDAC after one prior line of chemotherapy; ~92% RAS G12, KRAS G12D included in the G12 stratum
- **n:** 500
- **Toxicities:** any treatment-related AE (grade ≥3): 43.6%; rash (grade ≥3): 14%; stomatitis (grade ≥3): 12%; treatment discontinuation due to TRAE (grade any): 1.2%
- **Efficacy:** OS 13.2 months; HR_OS 0.4 (95% CI 0.3-0.53); HR_OS 0.4 (95% CI 0.3-0.54); OS_rate 53.2%; PFS 7.2 months; HR_PFS 0.49; ORR 31.6%; AE_rate 43.6%; discontinuation_rate 1.2%

Arid1A Loss interventions

1. ceralasertib (AZD6738), ATR inhibitor (ARID1A synthetic-lethal axis) — via NCT03682289, gated on BAF250a IHC loss

Likelihood of effect: Low for PDAC — confirmed ORR 14% across ARID1A-deficient tumors (pmid:41686845), activity concentrated in gynecologic histologies with zero pancreatic responders shown.

Toxicity burden: Moderate (ATR-class anemia / thrombocytopenia / neutropenia; per-term G3+ rates unreported, zero treatment-related deaths)

Counter-productive MoA: Moderate (Cross-tumor extrapolation with no PDAC responder; the synthetic-lethal effect may not engage in basal-like PDAC, and class anemia limits exposure at 79.)

Overall: A mechanistically independent banked second axis off the ARID1A frameshift, but no PDAC responder exists, entry is BAF250a-IHC-gated, and the lead basket is not recruiting.

Key references: PMID 41686845 · NCT03682289 · PMID 27958275

Evidence in detail

ceralasertib (AZD6738), ATR inhibitor (ARID1A synthetic-lethal axis) —

****Zhu et al. *Clin Cancer Res* 2026****

- **Disease context:** ARID1A-deficient advanced solid tumors (gynecologic-predominant) (2L+) — 29 evaluable, ARID1A loss by IHC required; mostly gynecologic (endometrioid endometrial, ovarian clear cell); non-gynecologic tumors under-represented
- **n:** 29
- **Toxicities:** treatment-related death (grade 5): 0%
- **Efficacy:** ORR 14%; ORR 31%; DoR 33.7 months